Faithful, Simplified, or Condensed? Designing and Testing Generation Modes for LLM-Generated Instructional Text

Abstract

Instructors increasingly use large language models (LLMs) to turn source documents into interactive course readings, but few such pipelines are designed with explicit, testable targets or evaluated against them. We present three generation modes for an intelligent textbook platform, each with a written specification: Faithful (verbatim preservation), Simplified (Grade 9–10 target), and Condensed (≈60% length). We then ask whether the generated texts actually differ as specified, using two methods. First, we conduct a blinded rating study in which 12 raters read 36 generated texts (12 source documents × 3 modes) on the live platform and rate cohesion, complexity, readability, and overall appropriateness on 6-point scales; we then test whether an LLM judge given the same instrument can reproduce those ratings. Mixed-effects models show Simplified and Condensed texts are rated more readable than Faithful ones, Condensed texts more cohesive, and Simplified texts less complex. Raters identified the mode above chance (46% vs. 33%) but far from reliably ($\kappa = .19$), with Condensed most often mistaken for Faithful. The LLM judge reproduced every human mode effect, agreed with individual human raters about as well as they agreed with each other, was far more consistent with itself than any human rater (single-pass ICC .67–.89 vs. .20–.41), and identified the mode in 92% of texts. Second, we calculated automated cohesion, lexical, and syntactic indices (TAACO, TAALES, TAASSC) and readability formulas over 300 texts from 100 source documents. These indices confirm that Simplified text is lexically and syntactically simpler, but reveal that Condensed text is more lexically sophisticated and less locally cohesive than Faithful text, the opposite of how readers perceived it, and that neither mode fully meets its numeric specification. We also report the blinded preference study that selected a validation-checklist prompting strategy for the pipeline. The results argue for specifying generation targets explicitly and checking them with readers, an LLM judge, and automated indices, since each catches what the others miss.

1 Introduction

As large language models improve, they are increasingly likely to be used to produce instructional text: course readings generated from a source document, rewritten for a particular audience, or shortened for review. Where this already happens, much of it is informal. A prompt is written, the output looks reasonable, and it is shipped. Far less common is the practice of writing down what the output is supposed to be, in measurable terms, and then checking whether it is. This paper is an exercise in that practice, carried out on a deployed intelligent textbook platform, extending a tournament-style approach to prompt evaluation for educational applications (Anonymous, 2025) from question generation to whole-document generation.

The platform, the Anonymized Intelligent Textbook Platform (AITP), is a web application in which an instructor or instructional designer uploads a source document and receives an interactive course reading: the text is divided into pages and short chunks, each chunk ends with an automatically generated question that must be answered before the next chunk unlocks, and learners' answers, reading time, and annotations flow to an instructor dashboard (Anonymous, 2025). Instructors who build courses on the platform regularly need the same source rendered at different levels of fidelity: an exact version for an advanced

course, an easier version for struggling readers, a shorter version for review. The choice is the instructor's, not the learner's; the modes are an authoring control, not a personalization mechanism.

We know from prior work that such variants should matter. Simplified text improves comprehension for English language learners (Crossley et al., 2014), and for native speakers it improves fluency for weaker readers and comprehension for readers with less background knowledge, though not for stronger or better-informed readers (Gruteke Klein et al., 2025). Condensing a document, by contrast, has been found to leave comprehension performance unchanged (Morris et al., 1992). What we do not know is whether an LLM, instructed to produce such a variant from an arbitrary source, actually does so: whether a "simplified" output is simpler in the ways that matter to readers and on the indices that predict comprehension, and whether a "condensed" output is merely shorter or also different in character. We therefore specified three generation modes with explicit, checkable targets (§3.1) and asked:

RQ1 Do readers perceive texts generated in different modes as differing in cohesion, complexity, readability, and overall appropriateness, and can they tell the modes apart? And can an LLM judge, given the same instrument, reproduce those human judgments reliably enough to stand in for them?

RQ2 Do the modes differ on automated discourse indices (cohesion, lexical sophistication, syntactic complexity, readability) in the directions their specifications intend?

Our contributions are: (i) the specification and implementation of the three modes, including a blinded comparison of three prompting strategies for the underlying pipeline (§3.1); (ii) a blinded, platform-based rating study of 36 generated texts by 12 raters with mixed-effects analysis and a mode-identification task, which serves as a gold-standard rating set (§3.3, §4.1); (iii) an LLM-as-judge replication of that study against the gold ratings, with reliability corrected for aggregation so that judge and humans are compared on equal terms (§3.3, §4.1); and (iv) an automated characterization of 300 generated texts using three established discourse-analysis tools and standard readability formulas (§3.4, §4.2).

2 Related Work

2.1 Intelligent Textbooks

Intelligent tutoring systems adapt instruction to learner state (VanLehn, 2011; Brusilovsky and Millán, 2007). Intelligent textbooks apply the same idea to reading: a static text is augmented with embedded questions, feedback, and analytics so that reading becomes an instrumented, interactive activity. LLMs have lowered the cost of building such textbooks by automating question generation, answer scoring, and feedback (Kasneci et al., 2023), and the platform studied here is one such system (§2.2). No prior work on intelligent textbooks, to our knowledge, has exposed explicit generation modes to authors or asked whether the modes yield reader-perceptible differences.

2.2 The Platform

AITP is a framework for interactive digital textbooks. It converts a source PDF into a hierarchical volume of pages, each containing one or more chunks. Each chunk holds body text (Markdown), inline figures extracted from the source, and a constructed-response item (CRI) consisting of a Question, a model ConstructedResponse, and 3–5 KeyPhrases. Readers must answer the CRI before the next chunk unlocks; responses are scored automatically and receive formative feedback (Anonymous, 2025). A teacher dashboard exposes focus time, annotation behavior, and assessment scores. The platform has shown learning gains over static textbooks in randomized trials (Anonymous, 2025) and improvements in reading and writing performance through AI feedback (Anonymous, 2025), and has been deployed in undergraduate computer science and psychology courses (Anonymous, 2025). Question generation and CRI scoring have been evaluated separately (Anonymous, 2025); the text itself has not.

2.3 Text Simplification and Condensation

Automatic simplification has been studied at the lexical, syntactic, and discourse levels (Siddharthan, 2014; Alva-Manchego et al., 2020), with long-standing concern about what "simpler" means and for whom (Xu et al., 2015). Its effects on readers are not uniform: simplification helps English language learners (Crossley et al., 2014) and, among native speakers, helps weaker readers read more fluently and low-knowledge readers comprehend more, with no benefit to stronger or better-informed readers (Gruteke Klein et al., 2025). LLMs now approach human quality on sentence-level simplification (Kew et al., 2023), but document-level simplification, where the effect must be measured in discourse terms, is less well characterized.

Condensation is closer to summarization (Nenkova and McKeown, 2011) than to simplification: it removes material rather than rewording it. An early controlled experiment found no difference in comprehension between automatically condensed documents and their originals (Morris et al., 1992). Because cohesive bridging material is a common casualty of compression, and cohesion interacts with prior knowledge in learning from text (McNamara et al., 1996), condensation is not automatically beneficial and may be harmful for readers who need the bridges.

2.4 Automated Discourse Analysis

Several tools compute large sets of linguistic indices from text. TAACO (Crossley et al., 2016) computes over 150 indices related to local and global cohesion, including lexical and semantic overlap at the sentence and paragraph levels and the incidence of connectives. TAALES (Kyle and Crossley, 2015) computes indices of lexical sophistication such as word frequency, range, academic register, age of acquisition, and concreteness. TAASSC (Kyle, 2016; Kyle and Crossley, 2018) computes fine-grained clausal and phrasal indices of syntactic complexity. ARTE (Choi and Crossley, 2022) computes classic and crowdsourced readability formulas (Crossley et al., 2017). None of these tools produces a single "cohesion" or "complexity" score; each produces many indices from which the analyst selects. These tools, and Coh-Metrix before them (Graesser et al., 2011), have been used to characterize registers (Biber and Conrad, 2009) and to predict comprehension (Graesser et al., 2011; Crossley et al., 2017). We use selected indices as objective counterparts to the constructs our raters judged.

2.5 Prompting Strategies for Structured Generation

Generating a textbook volume is a structured-output task: the model must emit valid JSON with pages, chunks, and CRIs that satisfy length, reading-level, and content rules. Three families of prompting strategy are candidates. Chain-of-thought prompting (Wei et al., 2022) asks the model to reason step by step before answering, which improves performance on multi-step tasks but gives no guarantee that the final output satisfies the stated constraints. Few-shot prompting (Brown et al., 2020) supplies worked input–output examples so the model can imitate the target format; it is effective for format adherence but encodes rules only implicitly, through the examples. Self-checking strategies such as Self-Refine (Madaan et al., 2023) and Chain-of-Verification (Dhuliawala et al., 2024) ask the model to audit its own draft against explicit criteria and revise, which makes the rules first-class and is well suited to schema- and rule-constrained generation. We compare the three families empirically (§3.1.1).

2.6 The Current Study

We add three generation modes to the platform's pipeline, Faithful, Simplified, and Condensed, each defined by a written specification (§3.1), and test whether the specifications are realized. RQ1 asks readers, on the live platform, and then asks whether an LLM judge can reproduce what the readers said. RQ2 asks the automated indices, on a larger corpus. The two sources of evidence turn out to disagree in an instructive way.

3 Methods

3.1 Generation

Pipeline. A volume is generated in a single call to Gemini 2.5 Flash via the OpenRouter API at the provider's default tempera-

ture. The model receives the source PDF itself as a file attachment, so it sees layout, tables, and figures rather than pre-extracted text, together with a prompt assembled from three parts: the base validation checklist shared by all modes (§3.1.1), which states the volume, page, and chunk schema and the chunking, Markdown, image, and content-quality rules as individually checkable items; a mode module, which adds the mode's own rules and a mode-specific version of the workflow below; and one complete reference volume as a worked example of the output schema. Before the call, PyMuPDF extracts each figure in the PDF with its page number, position on the page, and a caption identified heuristically from the surrounding text; this metadata is appended to the prompt so that the model places figures by identifier rather than describing them. The prompt requires the response to be the volume JSON alone (title, description, and summary; pages; chunks with a header, Markdown text, and, for interactive chunks, a question, a reference answer, and key phrases), which is parsed and uploaded to the CMS.

Within that call the prompt instructs the model to proceed through six stages. Extract the text and figure metadata. Transform the text according to the mode: identity for Faithful; sentence-level rewriting for Simplified (replace jargon, split long sentences, define unavoidable technical terms inline, reduce passive voice); and content reduction for Condensed (drop extended examples, redundant explanations, and tangential history, keep one or two examples per concept, merge paragraphs that make the same point). Chunk the result by the source's own headings into as few pages as possible, with chunks at the mode's target size (Faithful 400–500 words, Simplified 200–400, Condensed 150–300). Generate a CRI for each interactive chunk, rotating question types through conceptual, application, analysis, synthesis, and evaluation and increasing in difficulty across the volume, with a reference answer locatable in the chunk and key phrases drawn verbatim from it. Place images at their identifiers with generated alt text. Finally, Validate: check the draft against every item of the checklist, including the mode's own rules (e.g., Faithful FM1 "all text matches the source exactly"; Simplified SM1 "Flesch–Kincaid grade 9–10"; Condensed CM1 "total length 55–65% of source"), fix any violation, re-check, and only then emit the JSON. The stages are therefore instructions the model follows inside one generation, not separate calls, and the checklist is self-applied: nothing outside the model verifies the output, which is what RQ2 tests.

	Faithful	Simplified	Condensed
Length target	100%	100%	≈60%
Reading level	Source	Grade 9–10	Source
Chunk size (words)	400–500	200–400	150–300
Chunks / page	2–4	3–6	4–8
CRI answer (sent.)	2–3	1–2	1–2
Text modification	None	Lex./synt.	Deletion

Table 1: Generation mode specifications, as enforced by each mode's validation checklist.

Modes. Table 1 summarizes the three modes; each adds its own rules to the validation checklist.

Faithful preserves source wording exactly: no paraphrase, simplification, or omission. Large chunks maintain discourse continuity.

Simplified targets a Flesch–Kincaid grade level of 9–10 by replacing technical vocabulary with plain equivalents (e.g., methodology → method), shortening sentences to 15–20 words, reducing passive voice, and defining unavoidable technical terms inline on first use. Content is preserved; only surface form changes.

Condensed reduces total length to 55–65% of the source by removing extended examples (keeping one or two per concept), redundant explanation, and tangential context, while always retaining learning objectives, core concepts, key findings, and at least one example per major concept. Smaller chunks yield more frequent CRI checkpoints.

3.1.1 Selecting a Prompting Strategy

Before building the modes we compared three prompt designs for the base (Faithful-like) pipeline, following the blinded tournament procedure of Anonymous (2025), on 8 source documents: two introductory communication chapters, two introductory psychology chapters, a public speaking chapter, a lexical-resources chapter, and two U.S. civics (USCIS) chapters.

Document	S1 CoT	S2 Few	S3 Val.
Communication 1	0	1	5
Communication 2	0	2	4
Psychology 1	0	2	4
Psychology 2	3	2	1
Public speaking	1	4	1
Lexical resources	5	1	0
USCIS 1	1	0	5
USCIS 2	1	0	5
Total	11	12	25
%	22.9	25.0	52.1

Table 2: Blinded preference votes by prompting strategy (6 raters × 8 documents; S1 chain-of-thought, S2 few-shot, S3 validation).

- S1 Chain-of-thought: reason step by step through extraction, chunking, and CRI writing before emitting JSON.
- S2 Few-shot: three complete input–output examples of correct AITP structure (plain page, page with equations, page with images).
- S3 Validation: the rule checklist above; the model audits each field before finalizing.

Six raters (members of our research group, including graduate students in NLP and data science) saw the three outputs for each document in randomized, blinded order (labelled A/B/C), chose the best on chunking granularity, CRI completeness, faithfulness, and page structure, and wrote a justification. Table 2 shows the 48 votes.

S3 won 25/48 votes and 5/8 documents. Justifications were multi-coded into five themes: CRI field completeness (45.8% of comments; S1 often omitted KeyPhrases), chunk-length appropriateness (43.8%; S2 over-long, S1 over-short), adherence to chunking rules (37.5%), CRI quality (35.4%; S3 praised for depth and variety), and hallucinated content in page summaries (16.7%, present under all strategies). S3's checklist became the backbone of all three modes, with the mode-specific rules in Table 1 appended to it.

3.2 Corpus

We generated all three modes for 100 publicly available informational PDFs, yielding 300 texts. The corpus was built to cover ten knowledge domains (health and medicine; environment and climate; natural sciences; technology and computing; education; economics and finance; psychology; public policy and law; society and lifestyle; history). For each domain, an LLM proposed 50 candidate topics, ten were drawn at random, and each topic was combined with the descriptors fact sheet, overview, brief, guide, or chapter in a filetype:pdf Google search. The first result meeting all criteria was retained: 2–5 pages, text-heavy (≥200 words per page excluding cover and references), informational rather than promotional or form-like, selectable text, English, and publicly accessible. The final set was balanced to at least a 70/30 split of single- versus multi-column layouts and drawn from government, academic, and non-profit sources. For the automated analysis, the JSON volume produced by the generation model was converted to plain text: chunk headings and body text concatenated in reading order as a single paragraph, with no paragraph or chunk boundaries preserved. CRIs (questions, model answers, key phrases) were excluded, so the indices describe the expository text only. Because paragraph structure is not preserved, we report sentence-level cohesion indices only; paragraph-level cohesion is left to future work. Faithful volumes average 1,899 words (SD = 850, range 650–4,397).

3.3 RQ1: Human Rating Study

Materials. We sampled 12 source documents from the 100-document corpus (§3.2): one drawn at random from each of the ten domains, plus one more from each of two domains chosen at random. They span health (asthma; PTSD and post-traumatic growth; social media and mental health), science and technology (carbon capture; photosynthesis; 5G networks), social science and policy (land-grant institutions; consumer credit, two documents), law (precedent and jurisprudential disagreement), and history and political thought (the ancient Silk Road; a review of Athens on Trial). Each was generated in all three modes, giving 36 volumes, which were published to a private organization on the live AITP platform so that raters experienced the chunk-by-chunk interface, CRIs, and feedback exactly as students would.

Raters and design. Fifteen raters were recruited from our research group at a U.S. university: graduate students in NLP and data science and undergraduates from a range of majors; 14 enrolled and 12 completed the study (two returned no ratings; one omitted three volumes), yielding 141 ratings of the 36 volumes. Assignment used a rotating Latin square: each rater received 12 volumes, four per mode, and never saw two versions of the same document. Each of the 36 volumes was rated by 3–4 raters (mean 3.9). Raters were blind to mode and to the fact that documents recurred across raters. After data collection we verified each rated link against the volume it opened; four ratings had been issued a link to a different mode of the same document than assigned and were reassigned to the volume actually read, giving 49 Faithful, 46 Simplified, and 46 Condensed ratings. They were instructed to read and interact with each volume for at least 4–6 minutes before rating.

Instrument. Each volume received four 6-point ratings: Cohesion (Was the text well organized?, 1 = very disorganized, 6 = very organized); Complexity (How sophisticated and varied were the word choices and sentence structures?, 1 = very simple, 6 = very complex); Readability (Was the text easy to understand?, 1 = very difficult, 6 = very easy); and Overall (Was the text well formatted and appropriate for adult learners interested in this topic?, 1 = very ineffective, 6 = very effective). Unlike the other three scales, a higher Complexity rating is not a better one: it records the register the rater perceived, so that a Faithful volume of a technical source should score high and a Simplified one low. Raters then guessed the mode (given one-sentence definitions of the three modes) and could leave free-text comments on the content and on platform bugs.

Analysis. Each rating was modelled with a linear mixed model with mode as a fixed effect (Faithful as reference) and crossed random intercepts for source document and rater, fit by maximum likelihood; the omnibus effect of mode is a likelihood-ratio test against the intercept-only model. Mode identification is compared with chance (1/3) by an exact binomial test, and agreement with ground truth by Cohen's κ.

LLM-as-judge replication. To test whether a language model can stand in for human raters on this instrument, we gave an LLM judge (gpt-5.6-sol via the OpenAI Responses API; reasoning effort medium, temperature 1.0, top-p 0.98) the same task: for each of the 36 volumes, rate Cohesion, Complexity, Readability, and Overall on the same 6-point anchored scales, give a short explanation with evidence from the text, and guess the mode from the same one-sentence mode definitions. The judge received the volume (headings, chunk text, and CRIs) but, like the human raters, not the source document or the mode. Each volume was judged in six independent passes (two runs of three, one week apart, with identical settings), giving 216 judgments.

Judge analysis. The judge is evaluated in three ways. Its reliability is the one-way ICC across passes, reported both for a single pass, ICC(1), which is comparable to a single human rater, and for the mean of the six passes, ICC(1,6), which is the score used in the comparisons below; the mean of several passes is necessarily more reliable than one, so the two must not be confused. Its agreement with humans is the Pearson correlation between judge and human volume means, which we treat as interval as we do for the ICC, with Spearman's ρ and quadratic weighted kappa (QWK) as supplementary ordinal measures, and a disattenuated correlation that divides r by the square root of the product of the two aggregates' ICCs. Because a mean of six passes is not a fair match for one human, we also correlate a single judge pass with each human rater on the volumes they share and compare that to the same statistic between pairs of humans. Its mode identification is scored as for humans, by majority vote over passes and per judgment. The mixed models of §3.3 are refit to the judge's ratings with pass-level rows and a random intercept for source document.

3.4 RQ2: Automated Discourse Analysis

Tools and indices. Texts were processed with three tools from the Suite of Automatic Linguistic Analysis Tools (SALAT),[1] summarized

[1] https://www.linguisticanalysistools.org/

Tool	Construct and indices used
TAACO 2.1.3 (Crossley et al., 2016)	Cohesion (4): adjacent-sentence argument overlap; adjacent-sentence LSA similarity; incidence of basic connectives; incidence of all connectives.
TAALES 2.8.1 (Kyle and Crossley, 2015)	Lexical sophistication (9): COCA news log frequency (all words); word frequency (Freq_N); COCA bigram association (ΔP); proportion of bigrams in the top 30k; Academic Word List coverage; age of acquisition (all words); concreteness (all words); contextual distinctiveness (content words); lexical decision reaction time (z).
TAASSC 1.3.8 (Kyle, 2016)	Syntactic complexity (3): dependents per clause; coordinating conjunctions per clause; clausal complements per clause.
Direct (textstat)	Length and readability (4): word count; mean sentence length; Flesch–Kincaid grade level; Flesch reading ease.

Table 3: Tools, versions, and the 20 indices analysed in RQ2. All indices are per-text values computed on the 300 generated texts.

in Table 3. The 16 SALAT indices were selected a priori to cover cohesion, lexical sophistication, and syntactic complexity with minimal redundancy. We add word count, mean sentence length, and the Flesch–Kincaid grade level and Flesch reading ease formulas computed directly from the texts, since the Simplified mode's specification is stated in Flesch–Kincaid terms.

Analysis. Because each source document appears in all three modes, texts are not independent. Each index is analysed with a linear mixed model with mode as fixed effect and a random intercept for source document; the omnibus effect of mode is a likelihood-ratio test against the intercept-only model, Holm-corrected across the 20 indices. Pairwise contrasts are paired t-tests on the 100 document-level differences, Holm-corrected within index, with Cohen's d_z as effect size. We pre-specified directional expectations from the mode definitions: Simplified $<$ Faithful on sentence length, grade level, age of acquisition, and academic vocabulary, and $>$ Faithful on word frequency and adjacent overlap (shorter sentences repeat key terms); Condensed $\approx$ Faithful on lexical and syntactic indices but $<$ Faithful on adjacent-sentence overlap and similarity (bridging sentences removed).

	Faith. $n = 49$	Simp. $n = 46$	Cond. $n = 46$	$\chi^2(2)$
Cohesion	4.49 (1.08)	4.74 (0.95)	5.02 (0.80)	8.22^{*}
Complexity	3.90 (1.33)	3.28 (1.41)	3.59 (1.31)	$5.32^{\dagger}$
Readability	4.31 (1.08)	4.87 (1.07)	4.70 (0.94)	9.33^{**}
Overall	4.37 (1.13)	4.80 (1.00)	4.61 (1.11)	$5.31^{\dagger}$

Table 4: Mean (SD) ratings by mode on 6-point scales and likelihood-ratio χ^2 for the effect of mode in the mixed model. $^{\dagger}p < .08$, $^{*}p < .05$, $^{**}p < .01$.

4 Results

4.1 RQ1: Human Ratings

Reliability. Because each volume was rated by a different subset of raters, we report Krippendorff's α (interval) for single-rater agreement and the one-way intraclass correlation for the mean of the $\approx$4 raters per volume, ICC(1,k). Single-rater agreement was low to moderate (α: Overall .20, Cohesion .28, Readability .31, Complexity .41), as is typical for untrained raters judging text quality; reliability of the volume means used in the analyses below was moderate (ICC(1,k): .50, .61, .64, .73 respectively). A Spearman–Brown estimate from the mean pairwise inter-rater correlation ($\bar{r}$ over the 18 rater pairs who shared volumes: .18, .33, .23, .25; $k = 3.9$) gives a consistent picture (.46, .66, .54, .56). Mode guesses agreed weakly across raters (nominal $\alpha = .17$).

Ratings. Table 4 and Figure 1 report descriptives and omnibus tests. Readability differed by mode ($\chi^2(2) = 9.33$, $p = .009$): relative to Faithful, Simplified volumes were rated +0.58 points more readable ($SE = 0.19$, $p = .002$) and Condensed volumes +0.40 ($SE = 0.19$, $p = .038$). Cohesion also differed ($\chi^2(2) = 8.22$, $p = .016$), driven by Condensed volumes (+0.54, $SE = 0.19$, $p = .004$); Simplified did not differ from Faithful (+0.26, $p = .17$). Overall appropriateness was marginal at the omnibus level ($\chi^2(2) = 5.31$, $p = .070$), with Simplified rated higher than Faithful (+0.47, $SE = 0.20$, $p = .020$) and Condensed not reliably so (+0.27, $p = .19$). Complexity was likewise marginal ($\chi^2(2) = 5.32$, $p = .070$), though Simplified volumes were rated less complex than Faithful (-0.62, $SE =$

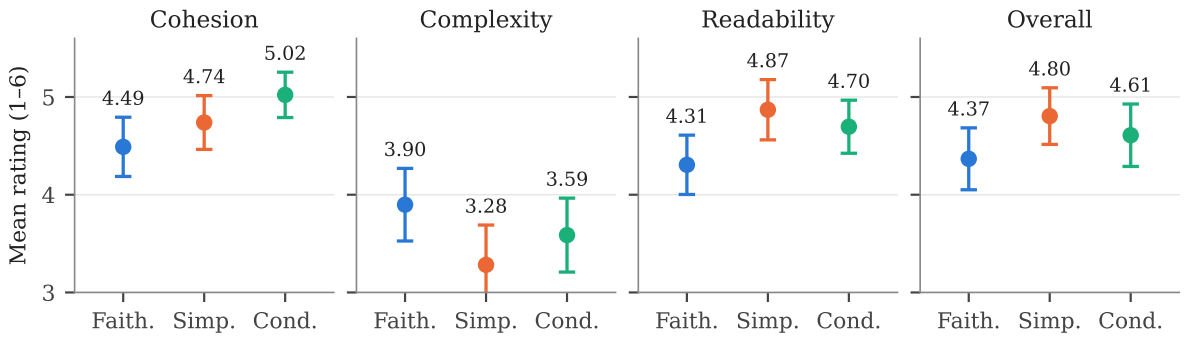


Figure 1: Mean rating by mode on each 6-point scale, with 95% confidence intervals ($n = 46$–49 ratings per mode). Simplified and Condensed volumes are rated more readable than Faithful; Condensed volumes more cohesive; Simplified volumes less complex.

	Guessed			
True	Faith.	Simp.	Cond.	Recall
Faithful	27	7	15	.55
Simplified	12	22	12	.48
Condensed	20	10	16	.35

Table 5: Human mode identification (141 guesses). Accuracy 46.1%, chance 33.3%; $\kappa = .19$.

0.26, $p = .020$). Complexity and readability were strongly inversely related across volumes (Spearman $\rho = -.64$, $p < .001$).

Mode identification. Raters identified the mode correctly in 65 of 141 cases (46.1%), above chance (binomial $p = .002$; $\chi^2(4) = 15.90$, $p = .003$) but with weak agreement ($\kappa = .19$). Table 5 shows the pattern: Faithful was easiest to recognize (recall .55), Simplified intermediate (.48), and Condensed hardest (.35). Condensed volumes were more often labelled Faithful (20/46) than Condensed (16/46). This is expected: raters never saw the source, and a well-executed condensation that keeps register and structure intact leaves no surface cue that anything was removed. Simplified was the most precise guess (.56): when raters said "simplified" they were usually right.

Comments. Raters left 52 content comments and 31 bug reports. Content comments clustered by mode. For Faithful, the recurring complaint was chunks rendered as single long paragraphs and source material that a textbook would drop (contact lists, reference sections, author biographies). For Condensed, nearly every rater flagged unexplained bold text on arbitrary words, alongside orphaned citation markers ("[12]") and missing spaces after sentence boundaries; one rater judged a Condensed volume "too simplified." For Simplified, comments were mostly positive about inline definitions and elaborations, with two raters noting monotonous sentence structure and meta-narration ("This section will…"). Bug reports were mode-independent: unrendered figures and LaTeX equations, Markdown tables and emphasis shown as raw syntax, and CRI scoring that accepted copied fragments of the question as correct. The last of these is a platform issue but affects how "interactive" volumes are perceived.

LLM judge. Table 6 summarizes. Three findings. First, the judge is far more consistent than a human rater. A single judge pass has ICC(1) .67–.89 and pass-to-pass QWK .66–.88, against ICC(1) .20–.41 and QWK .15–.28 for a single human rater; the mean of six passes used below has ICC(1,6) .92–.98, and the mode guess was identical in all six passes for 32 of 36 volumes. It also uses a narrower band of the scale (Cohesion ratings were 4–6 only; Overall was 5 in 154 of 216 judgments). Second, the judge reproduces every mode effect found with humans, with the same signs and larger magnitudes: relative to Faithful, Simplified was rated more readable ($+0.99$, $p < .001$), less complex (-1.39, $p < .001$), and higher overall ($+0.28$, $p < .001$); Condensed was rated more cohesive ($+0.36$, $p < .001$), more readable ($+0.71$, $p < .001$), and slightly less complex than Faithful (-0.22, $p = .001$), where humans found no difference. Agreement between

	Cohesion	Complexity	Readability	Overall
Judge means by mode (6 passes)				
Faithful	5.04	5.01	4.07	4.44
Simplified	5.33	3.62	5.06	4.72
Condensed	5.40	4.79	4.78	4.65
Reliability: single unit				
Judge, one pass: ICC(1)	.67	.89	.84	.87
Judge, pass vs. pass: mean QWK	.66	.88	.84	.87
Human, one rater: ICC(1)	.29	.41	.31	.20
Human, rater vs. rater: mean QWK	.28	.19	.20	.15
Reliability: aggregate used in analyses				
Judge, mean of 6 passes: ICC(1,6)	.92	.98	.97	.98
Humans, mean of ≈4 raters: ICC(1,k)	.61	.73	.64	.50
Judge–human agreement: aggregates ($N = 36$ volumes)				
r, judge mean vs. human mean	.32	.43**	.49**	.33*
ρ, judge mean vs. human mean	.27	.45**	.59**	.36*
r, disattenuated	.43	.51	.62	.48
QWK, judge mean vs. human mean	.12	.20	.42	.28
Judge–human agreement: judge as one more rater				
mean r, one judge pass vs. each human rater	.21	.38	.32	.21
mean r, human vs. human (ref.)	.33	.25	.23	.18
mean QWK, one judge pass vs. each human rater	.14	.29	.26	.18
mean QWK, human vs. human (ref.)	.28	.19	.20	.15

Table 6: LLM judge replication. ICC(1): reliability of a single pass or rater; ICC(1,k): reliability of the mean of k passes or raters, which is the score the analyses use. QWK: quadratic weighted kappa, mean over pairs of passes or raters sharing ≥ 6 volumes. r: Pearson correlation; ρ: Spearman correlation ($^{*}p < .05$, $^{**}p < .01$); the disattenuated r divides by $\sqrt{\mathrm{ICC}_{\mathrm{judge}}\,\mathrm{ICC}_{\mathrm{human}}}$ of the two aggregates. The last block compares one judge pass against each human rater to the humans' agreement with each other, so that both sides are single units.

judge and human volume means was moderate ($r = .32$–$.49$; .43–.62 after disattenuation for the unreliability of both aggregates), and a single judge pass agreed with individual human raters about as well as the humans agreed with each other ($\bar{r} = .21$–$.38$ vs. .18–.33; QWK .14–.29 vs. .15–.28). Third, the judge identified the mode in 33 of 36 volumes by majority vote over the six passes (92%, $\kappa = .88$; 196 of 216 individual judgments, 91%), against 46% for humans. Every error, in every pass, was a Condensed volume: Faithful and Simplified were identified perfectly, while Condensed was called Faithful or Simplified in 20 of 72 judgments, the same confusion humans made most often. The judge and the human majority agreed on the mode for only 56% of volumes, because the humans were usually wrong. A model reading the full text with unlimited attention usually detects condensation that readers on the platform do not, which suggests the cue is distributional (density of technical vocabulary, absence of examples) rather than anything a reader notices sentence by sentence.

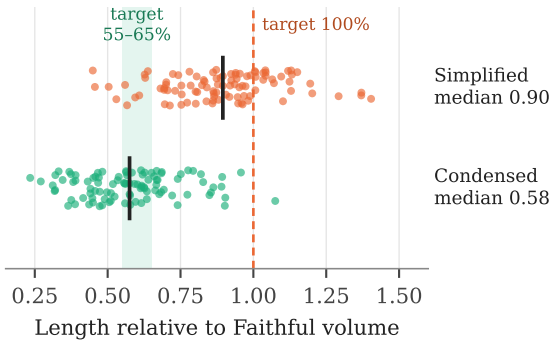


Figure 2: Length of each Simplified and Condensed volume relative to the Faithful volume of the same source ($N = 100$ each). Black bars mark medians; shaded band and dashed line mark the specified targets.

4.2 RQ2: Automated Indices

Specification compliance. Before the linguistic indices, two direct checks (Figure 2). Condensed volumes averaged 59% of the corresponding Faithful volume's length (median 58%), on target, but with high variance ($SD = 0.22$): only 27 of 100 fell inside the specified 55–65% band, 39 were shorter, and 34 were longer. Simplified volumes, specified at

100% length, averaged 91% ($SD = 0.24$), so simplification removed roughly a tenth of the text even though the mode forbids omission. Both results indicate that the length rules in the validation checklist are enforced only approximately by the model. The reading-level rule fared worse: Simplified volumes averaged Flesch–Kincaid grade 11.9 ($SD = 1.7$) against a 9–10 target, down from 13.6 ($SD = 2.6$) for Faithful; only 22 of 100 fell in the target band and 72 were above it. The mode moves text about two grade levels toward the target without reaching it.

Table 7 reports all indices. Mode had a significant omnibus effect on 18 of 20 after Holm correction; the exceptions were adjacent-sentence LSA similarity and the all-connectives index.

Simplified vs. Faithful. Every expectation for Simplified held. Sentences were shorter (16.7 vs. 20.0 words, $d_z = -0.85$), grade level lower ($d_z = -0.99$), and reading ease higher ($d_z = 0.89$). Vocabulary was acquired earlier ($d_z = -0.77$), more concrete ($d_z = 0.67$), faster in lexical decision ($d_z = -0.93$), less academic ($d_z = -0.64$), and less contextually distinctive ($d_z = -1.11$). Syntactically, Simplified text had more clausal complements per clause ($d_z = 0.65$), consistent with the inline definitions and explanatory "that"-clauses raters noticed. One expectation failed: adjacent-sentence argument overlap did not increase ($d_z = -0.01$, $p = .96$), and basic connectives decreased ($d_z = -0.49$). Simplification changed words and sentences, not the cohesive ties between them.

Condensed vs. Faithful. Condensed departed from its specification in the opposite direction. Rather than preserving the source's lexical profile, it became markedly more sophisticated: higher age of acquisition ($d_z = 2.37$), more Academic Word List coverage ($d_z = 1.62$), lower word frequency ($d_z = -2.23$ on COCA log frequency), higher contextual distinctiveness ($d_z = 1.79$), and slower lexical decision ($d_z = 2.12$). Grade level rose from 13.6 to 14.7 ($d_z = 0.56$) and reading ease fell from 32 to 20 ($d_z = -1.50$), even though sentences were shorter ($d_z = -0.67$). The mechanism is visible in what condensation removes: examples, restatements, and connective tissue are high-frequency, concrete text, so deleting them concentrates the technical vocabulary that remains. The cohesion prediction held: adjacent-sentence argument overlap fell ($d_z = -0.87$), and clausal complements and coordinating conjunctions per clause both decreased.

Simplified vs. Condensed. The two modes are furthest apart on every lexical index ($|d_z|$ from 0.8 to 3.1), and Condensed is below Simplified on argument overlap ($d_z = -0.90$). On the automated profile, Condensed is the hardest of the three texts to read and Simplified the easiest, with Faithful between them.

5 Discussion

Modes produce perceptible differences, but not the ones readers name. Readers rated Simplified and Condensed volumes as more readable than Faithful ones and Condensed volumes as more cohesive, yet they could attribute a volume to its mode only weakly. The two findings are compatible: what readers perceive is the effect of a transformation (easier, tighter), not the transformation itself. For platform design this means mode labels should be chosen by instructors who know the source, not inferred by students from the text.

An LLM judge is a usable proxy, with caveats. The judge reproduced the human mode effects with six passes on 36 volumes, at a reliability no feasible human panel would reach, and its agreement with individual humans matched inter-human agreement. For screening generated text at scale, that is sufficient. It is not a replacement for human judgment on the Overall scale, where the judge gave 5 to three-quarters of volumes and correlated least with humans, nor on anything the humans experienced through the platform (CRI scoring, navigation, rendering bugs) that the judge never saw. Its near-perfect mode identification is also a reminder that "can the reader tell?" and "can a model tell?" are different questions; only the first bears on learner experience.

Condensation reads as better organization. The strongest cohesion effect belonged to Condensed, not Simplified. Removing tangents and redundant examples appears to make the remaining structure more visible. The au-

Index	Faithful	Simplified	Condensed	$\chi^2(2)$	Pairwise
Length					
Word count	1963.94 (899.94)	1679.75 (648.90)	1096.21 (482.57)	161.9^{***}	F≠S, F≠C, S≠C
Mean sentence length (words)	20.02 (4.73)	16.71 (2.36)	17.24 (3.07)	75.7^{***}	F≠S, F≠C
Readability					
Flesch–Kincaid grade level	13.64 (2.56)	11.87 (1.74)	14.72 (2.00)	166.8^{***}	F≠S, F≠C, S≠C
Flesch reading ease	32.45 (12.79)	39.36 (10.29)	20.01 (11.10)	278.3^{***}	F≠S, F≠C, S≠C
Syntax					
Dependents per clause	0.12 (0.06)	0.14 (0.07)	0.15 (0.07)	17.0^{***}	F≠S, F≠C
Coordinating conj. per clause	0.003 (0.005)	0.002 (0.003)	0.001 (0.003)	16.8^{***}	F≠S, F≠C, S≠C
Clausal complements per clause	0.07 (0.03)	0.09 (0.04)	0.06 (0.03)	79.8^{***}	F≠S, F≠C, S≠C
Cohesion					
Adjacent-sentence argument overlap (2)	0.22 (0.07)	0.22 (0.07)	0.18 (0.06)	87.0^{***}	F≠C, S≠C
Adjacent-sentence LSA similarity (2)	0.72 (0.06)	0.72 (0.06)	0.71 (0.06)	5.7	S≠C
Basic connectives	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	31.9^{***}	F≠S, S≠C
All connectives	0.07 (0.01)	0.07 (0.01)	0.07 (0.01)	2.9	–
Lexical					
COCA news log frequency (AW)	2.56 (0.16)	2.56 (0.12)	2.32 (0.14)	320.0^{***}	F≠C, S≠C
Word frequency (Freq_N)	8.41 (0.19)	8.27 (0.16)	8.15 (0.22)	186.9^{***}	F≠S, F≠C, S≠C
COCA bigram association (DP)	0.07 (0.010)	0.06 (0.008)	0.07 (0.01)	138.3^{***}	F≠S, F≠C, S≠C
Bigram proportion in top 30k	0.31 (0.06)	0.28 (0.05)	0.20 (0.05)	299.6^{***}	F≠S, F≠C, S≠C
Academic Word List coverage	0.11 (0.04)	0.10 (0.03)	0.14 (0.04)	266.8^{***}	F≠S, F≠C, S≠C
Age of acquisition (AW)	6.48 (0.43)	6.28 (0.35)	6.98 (0.37)	354.4^{***}	F≠S, F≠C, S≠C
Concreteness (AW)	2.46 (0.13)	2.50 (0.11)	2.48 (0.12)	30.3^{***}	F≠S, F≠C, S≠C
Contextual distinctiveness (CW)	1.02 (0.11)	0.94 (0.09)	1.13 (0.10)	325.0^{***}	F≠S, F≠C, S≠C
Lexical decision RT (z)	-0.50 (0.03)	-0.52 (0.03)	-0.46 (0.03)	347.8^{***}	F≠S, F≠C, S≠C

Table 7: Automated indices by mode, mean (SD) over 100 documents. χ^2: likelihood-ratio test for mode in a mixed model with document random intercept, Holm-corrected across indices ($^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$). Pairwise: contrasts significant at Holm-corrected $p < .05$ (F = Faithful, S = Simplified, C = Condensed).

tomated indices say otherwise: Condensed text has the lowest adjacent-sentence argument overlap of the three modes and the most difficult vocabulary, and it scores a full grade level above Faithful on Flesch–Kincaid. Readers who rated it more cohesive and more readable were, we suggest, responding to the absence of digression and to shorter chunks with more frequent checkpoints, not to the linguistic properties that cohesion and readability indices measure. The two kinds of evidence are not in conflict so much as about different things: perceived organization versus measured lexical and referential ties. For learners with low prior knowledge, the measured profile is the one that predicts comprehension (McNamara et al., 1996), so Condensed mode as specified may be less suitable for them than raters' judgments imply.

Simplification lowers perceived complexity without lowering perceived quality. Simplified volumes were rated less complex, more readable, and higher overall. In this adult-reader sample, at least, simplification did not register as "dumbing down." The automated profile confirms the cost: Simplified text has lower academic vocabulary coverage and earlier-acquired, more concrete words. Whether this matters depends on the course; for a review text the loss is acceptable, for a text that teaches the technical register it is not. The profile also shows the mode overshoots its specification in one direction (it drops 9% of the words despite a no-omission rule) and undershoots in another (grade 11.9 against a 9–10 target).

Hallucination. Page-summary hallucination appeared under every prompting strategy (§3.1.1) and, in the mode study, one rater could not locate the answer to a CRI in its chunk, and another found a Condensed chunk that answered its own question verbatim. This mirrors the general NLG literature (Ji et al., 2023); the validation checklist reduces structural errors but does not verify content against the source. A source-grounded verification pass (Dhuliawala et al., 2024) is a natural next step.

Limitations. (1) Twelve raters and 36 volumes give adequate power for main effects but not for mode × domain interactions, and single-rater reliability was low; conclusions rest on volume means, not individual judgments. (2) Raters were research-group mem-

bers and university students, not the broader learner populations the Simplified and Condensed modes target. (3) Raters did not see the source, so faithfulness and coverage were not rated; the automated length and overlap indices are the only check on what Condensed removed. (4) The exported texts lack paragraph boundaries, so only sentence-level cohesion is measured; chunk- and paragraph-level cohesion, which the modes' chunking rules target directly, remain untested. (5) The 100-document corpus covers general-interest domains and omits highly technical registers (medicine, law) where mode trade-offs may differ. (6) Outputs are stochastic; all volumes were generated once with Gemini 2.5 Flash at default temperature and not regenerated. (7) The prompting-strategy study used a single base mode; strategies were not recompared per mode. (8) The judge study used one model and one prompt; the volumes were supplied as JSON files whose names carry a numeric suffix that correlates with generation order, and we have not ruled out that the judge used this cue in mode identification. The rating results do not depend on it.

6 Conclusion

We specified three generation modes for an intelligent textbook platform and tested whether they matter to readers. In a blinded study on the live platform, Simplified and Condensed volumes were rated more readable than Faithful ones, Condensed volumes more cohesive, and Simplified volumes less complex and more appropriate overall; readers could identify the mode only weakly, and rarely detected condensation. Automated discourse indices over 300 texts confirm that Simplified text is lexically and syntactically simpler, but show that Condensed text is more lexically sophisticated and less locally cohesive than the Faithful text it was cut from, and that neither mode fully meets its numeric specification. An LLM judge given the human instrument reproduced the human mode effects and identified modes that readers could not. Together the results support exposing mode-level controls to instructors while cautioning that readers cannot be expected to know which mode they are reading. Future work will test learning outcomes by mode in classroom A/B deployments and add source-grounded verification to the pipeline.

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